

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

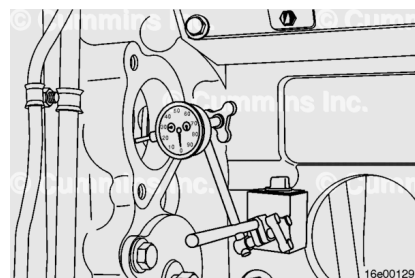
- Disconnect the batteries. See equipment manufacturer service information.
- Disconnect the electrical connections from the starting motor. Refer to Procedure 013-020 in Section 13. (</qs3/pubsys2/xml/en/procedures/57/57-013-020-tr.html>)
- Remove the transmission, clutch, and all related components. See equipment manufacturer service information.

Deflection of the flexplate **must** be measured when the generator is decoupled in order to check the flexplate suitability for reuse.

Before removing the generator from the engine, position a dial test indicator with peak retaining indication to measure maximum displacement of the flexplate at the moment of generator decoupling.

Record the measured value.

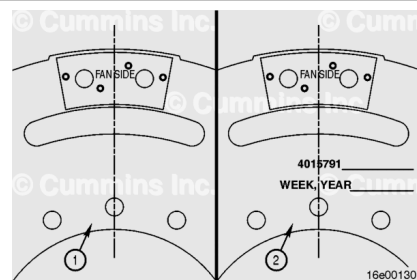
If deflection measurement can **not** be taken due to lack of suitable dial test indicator or access restrictions in vehicle, replace the flexplate with new one.



Displacement limits for reuse of the flexplate are:

- Previous generation flexplates (1) - 5.6 mm (0.22 inch).
- New generation flexplates (2) - 10.0 mm (0.39 inch).
- Insulated flexplates do **not** have displacement limits.

Note : On new generation flexplates (2), part number and production week/year are etched centrally under the stamped or etched marking indicating “FAN SIDE” located close to outer edge of the flexplate. Correctly installed flexplates will have the “FAN SIDE” marking,



facing outward from the engine block toward the generator fan.

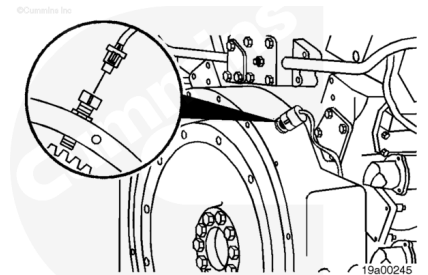
If the flexplate deflection is greater than the limit, the flexplate **must** be replaced.

Remove

Disconnect the sensor from the engine wiring harness.

Loosen the locking nut.

Remove the engine speed sensor from engine flywheel housing.

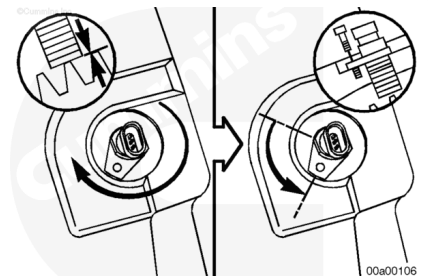


Remove the engine speed sensor (power generation).

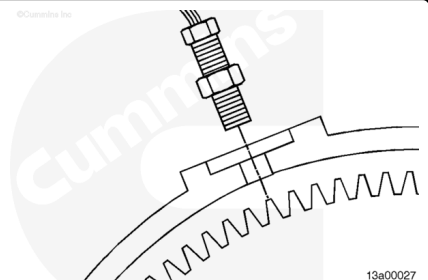
Note : This sensor can also be used in some industrial applications.

Disconnect the sensor from the engine harness.

Loosen the jam nut. Back the sensor out of the flexplate housing.



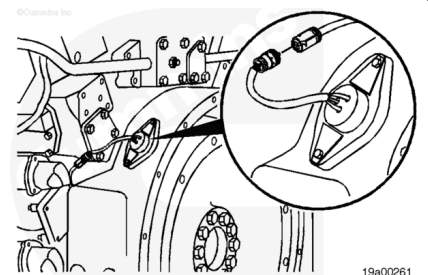
Remove the sensor.



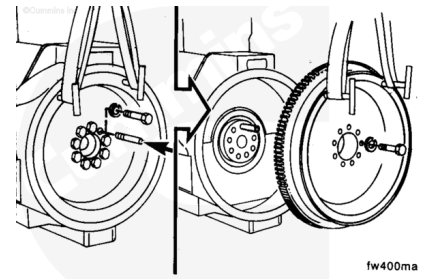
Disconnect the engine position sensor.

Remove the two capscrews securing the engine position sensor (EPS) capsule to the flexplate housing.

Remove the capsule, sensor, and harness connector as an assembly.



This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Remove two capscrews and install the guide studs.

Use two M20 guide studs to prevent the flexplate from rotating.

Use a hoist, two [5/8-11] T-handles, and a lifting sling. Install the T-handles.

Remove the remaining capscrews.

Remove the flexplate.

Note : Use a mallet to tap the flexplate from the crankshaft, if necessary.

Note : Industrial engines use an alignment dowel pin to align the crankshaft and flexplate properly. The dowel is a loose fit, not an interference fit. It can drop out when the flexplate is removed.

Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

Do not disassemble the flexplate assembly. Improper assembly of the flexplate can cause engine damage. Refer to Procedure 016-999 in Section F.
(/qs3/pubsys2/xml/en/procedures/57/57-016-999-tr.html)

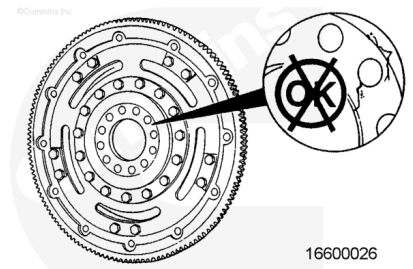
Clean the flexplate with solvent and dry with compressed air.

Check the ring gear for damaged or worn teeth. If the ring gear is damaged or worn, the assembly **must** be replaced.

Check the mounting pilot for damage. If the pilot is damaged, the flexplate **must** be replaced.

Use the dye penetrant method to check the area of the mounting holes for cracks. There **must not** be any cracks in the flexplate mounting area.

Check the mounting pilot for damage. If the pilot is damaged, the flexplate **must** be replaced.



Check the tightness of the twenty-five capscrews.

Torque Value:

Flexplate capscrews 271 n·m [200 ft-lb]

If replacement is required, use **only** the correct grade 10.9 capscrews released by Cummins Inc.

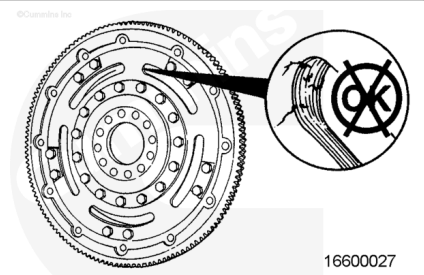
If the capscrews require replacement, remove and replace them one at a time. Do **not** disassemble the flexplate.



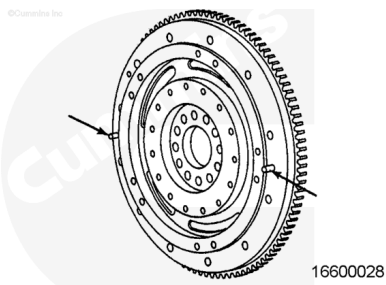
The flexplate can consist of as many as nine separate steel plates.

Check closely for cracks.

If any of the steel plates are cracked or otherwise damaged, the engine flexplate **must** be replaced.



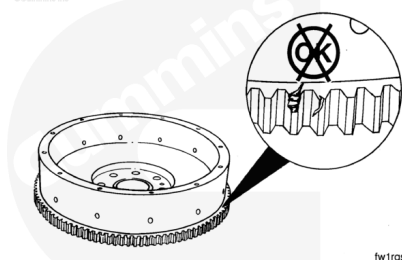
Check the two tapered dowels for damage. If the dowels are bent or broken, they **must** be replaced.



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Inspect the flexplate ring gear teeth for cracks and chips.

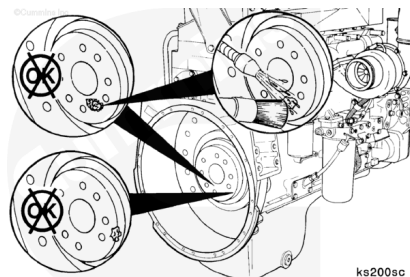
Note : If the ring gear teeth are cracked or broken, the ring gear must be replaced.



fw1rgsa

Inspect the crankshaft for dirt or damage.

Refer to Procedure 001-016 in Section 1 to replace the crankshaft. (/qs3/pubsys2/xml/en/procedures/57/57-001-016-tr.html)

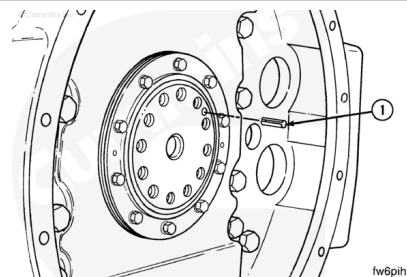


ks200sc

Install

Install the roll pin (1) in the crankshaft.

The pin is required to align the timing marks that are on the flexplate.



fw6piha

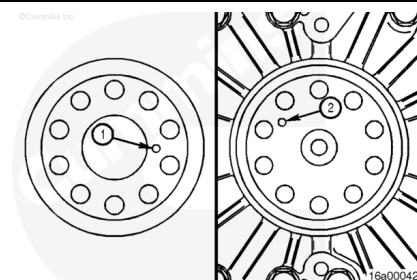
Lubricate the flexplate mounting capscrew threads and flange head surfaces with clean engine oil.



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⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.



The flexplate mounting capscrews are manufactured to a precise specification with rolled threads. If replacement is required, **only** the parts released by Cummins® can be used, otherwise a malfunction may result which could cause personal injury.

The use of a guide bolt will aid the assembly procedure.

Check to be sure the mating surfaces of the crankshaft and flexplate are clean, dry, and free of nicks or burrs.

Guide studs will help during assembly. Install guide studs at the 12-o'clock and 6-o'clock positions. Install the flexplate, washers, and capscrews. The flexplate **must** be firmly against the crankshaft.

Note : Industrial engines have a special M20 x 1.5 x 60 shouldered capscrew (1) that **must** be installed and tightened snug first into the countersunk hole in the flexplate. This capscrew requires no washer. A loose-fit dowel also **must** be installed into the flexplate.

Note : Some industrial engines will **not** have the dowel pin installed.

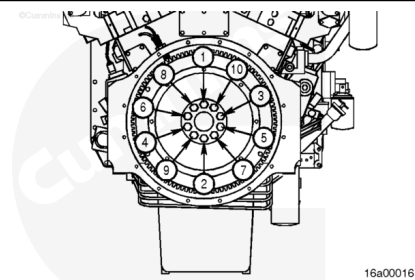
Note : The shouldered capscrew aligns the flexplate with the crankshaft. The dowel pins align the flexplate with the crankshaft for timing purposes.

Align the hole in the flexplate for the roll pin with the roll pin in the crankshaft.

Use the following steps to tighten the capscrews in the sequence shown.

Torque Value:

1. 102 n•m [75 ft-lb]
2. 298 n•m [220 ft-lb]
3. 542 n•m [400 ft-lb]



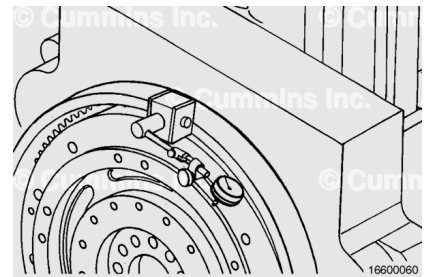
Measure

Note : The crankshaft **must** be pushed or pulled in the same direction each time a measurement is made in order to obtain an accurate reading.

Bar the engine from the front of the crankshaft for both radial and face measurements. The barring device will cause the flexplate to distort if barred from the rear, and will also cause the crankshaft to walk up the main bearing sidewall.

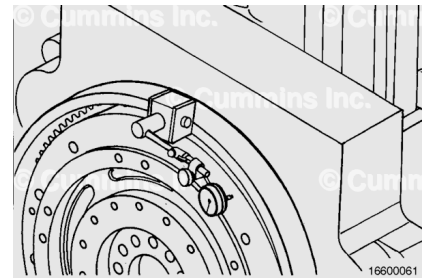
Radial Runout

- Attach an indicator to the flywheel housing with the indicator tip on the flexplate as illustrated. Check the indicator while rotating the engine.
 - Maximum Radial Runout is 0.38 mm [0.015 in]
- If the runout is **not** within specification, the pilot on the flexplate is **not** seated correctly on the crankshaft. If the pilot is damaged, the entire flexplate assembly **must** be replaced.



Face Runout

- Attach an indicator to the flywheel housing with the indicator tip on the face of the flexplate. Check the indicator while rotating the engine.
 - Maximum Face Runout is 0.38 mm [0.015 in]
- If the alignment is **not** within specification, check for foreign material between the flexplate and the crankshaft.
- Two 3/8 x 16 inch threaded holes are provided in the adapter to aid in the removal of the assembly.



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the transmission, clutch, and all related components. See equipment manufacturer service information.
- Connect the batteries. See equipment manufacturer service information.
- Connect the electrical connections to the starting motor. Refer to Procedure 013-020 in Section 13. (</qs3/pubsys2/xml/en/procedures/57/57-013-020-tr.html>)
- Start the engine. Check equipment operation.

Last Modified: 21-Feb-2023
